

Ureteric stent

A ureteric stent is a thin, flexible tube that can be placed in the ureter to hold it open. This allows urine (pee) to drain from the kidney into the bladder.

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What is a ureteric stent?

A ureteric stent is a thin, flexible tube that can be placed inside the ureter to hold it open. The ureters are the tubes that carry urine (pee) from each kidney to the bladder. This type of stent is also sometimes called a ureteral stent or kidney stent.

The top end of the stent sits in the kidney. The lower end sits in your bladder. Both ends of the stent are curved. This helps hold it in position.

If it is not possible to put in a stent, you may have a tube put into your kidney (or kidneys) through the skin on your back. This is called a [nephrostomy](#).

The urinary system

The urinary system is made up of the kidneys, the ureters, the bladder and the urethra.

The kidneys are at the back of the body, one on each side, just underneath the ribcage. The kidneys are bean-shaped. They are just above the level of the belly button. Kidneys filter your blood to get rid of waste products, and change them into urine (pee).

Urine travels from each kidney through a fine tube called a ureter to the bladder, where it is stored. The bladder is in the lower part of the tummy (abdomen). This area is called the pelvis. When you are ready to pass urine, it leaves the bladder through a tube called the urethra. The urethra opens in front of the vagina, or at the tip of the penis.

Image: The urinary system

Reasons for having a ureteric stent

You may need ureteric stents if cancer blocks one or both ureters. If a ureter is blocked, urine (pee) cannot flow from the kidney to the bladder. This causes urine to build up in the kidney. When this happens, the kidney may slowly stop working.

If a blockage is not treated quickly it can make you feel very unwell. You may:

- have pain or discomfort in your back or side
- have a high temperature (fever)
- feel sick.

Your doctor may suggest an operation to put a stent into one or both ureters.

Ureteric stent insertion

Ureteric stents can be put in in 2 different ways. Your doctor will explain what to expect.

Retrograde stenting

The stents are put in from inside the bladder while you are under general anaesthetic. They are placed using a cystoscope. This is a thin tube with a camera and light on the end.

The doctor passes the cystoscope through your urethra and into your bladder. They put the stent (or stents) into the ureters using a thin guide wire that they pass through the cystoscope.

Image: Cystoscopy

We have more information about having a [cystoscopy](#).

Antegrade stenting

The stents are put in through the back of your body, below your ribs. You have injections to numb the area first. Sometimes, people have drugs to make them sleepy and relaxed. This is called a sedative. The doctor uses a fine needle and thin guide wire to pass the stent into the kidney and down into the ureter.

After stent insertion

You will usually be able to go home the same day or the next day. Your doctor or nurse will give you advice about drinking plenty of fluids and any painkillers you may need.

They should also explain when the stent should be changed or removed. Ureteric stents can usually stay in for a number of months. Some can be left in for longer if needed.

The stent should let urine (pee) drain into the bladder and prevent damage to the kidney. But it also may cause side effects. The most common side effects are the following:

- Bladder symptoms – you may need to pass urine more often or more urgently. You may have small amounts of blood in your urine.
- [Pain](#) – you may have some pain or discomfort when you pass urine. Or feel pain in your lower back or groin. Pain may be worse after physical activity.
- Infection – you are more likely to get a urine infection. Symptoms include a high temperature, pain or difficulty passing urine, pain in your back or bladder, feeling unwell.

Drinking fluids can help prevent and may reduce side effects. Try to drink at least 2 litres (3½ pints) of fluids a day. Try to drink water instead of drinks that may irritate the bladder. These include:

- drinks containing caffeine, such as tea, coffee, drinking chocolate and cola
- alcohol
- fizzy drinks
- acidic drinks – for example, orange or grapefruit juice
- drinks with artificial sweeteners – for example, diet or 'light' drinks.

Always tell your doctor or nurse if you have these side effects. They can give you advice and treatment if needed.

About our information

References



Below is a sample of the sources used in our bladder cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Mottet N, Bellmunt J, Briers E, et al. Non-muscle-invasive bladder cancer (TaT1 and CIS). European Association of Urology (Internet), 2021. Available from uroweb.org/guideline/non-muscle-invasive-bladder-cancer (accessed September 2021).

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Ursula McGovern](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help